On a mixed GPU cluster, the slowest card sets the pace of every step

Illustrative arithmetic on datasheet peaks, not a measured result

On paper, an H100 does about three times the BF16 work of an A100

3.2x

peak dense BF16 throughput, H100 SXM against A100

About 990 against 312 TFLOPS. NVIDIA's headline 1,979 TFLOPS for H100 is with sparsity.

Source: NVIDIA H100 product page and A100 datasheet; H100 dense figure is half the with-sparsity peak

Split the batch evenly and the faster card waits at every synchronisation

68%

of each step an H100 sits idle, given the same share as an A100

1 minus 312/990. Memory bandwidth, interconnect and the model change the real ratio, so measure yours.

Source: Illustrative: arithmetic on NVIDIA datasheet peaks

With an even split, the slowest card still sets the step time

EVEN SPLIT

Same batch on every GPU

Step time set by the slowest card

Fast cards wait at the barrier

SPLIT BY MEASURED SPEED

Batch share sized to each card's measured throughput

Or group cards into pipeline stages, as HetPipe does

Re-measure after every change

Researchers called this years ago

> **For training of large DNN models, the use of heterogeneous GPUs is inevitable due to the short release cycle of new GPU architectures.**

Park et al., HetPipe, arXiv:2005.14038, 2020

How we would approach it: measure before optimising

The plan in our research collaboration with a Norwegian research laboratory, in progress

01 **Survey**

Map how multi-GPU, multi-node work is split today.

02 **Benchmark**

Build a benchmark for mixed GPU setups.

03 **Change one thing**

Propose optimisations and measure what each one actually changes.

How do you split work across GPU generations today?

Scoping is short, fixed and ends in a written plan.

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